

Replication / Machine Learning

# [Re] Machine Learning Assisted Cervical Cancer Detection: Reproduction and Evaluation Analysis

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**Abstract** We report a replication of Machine Learning Assisted Cervical Cancer Detection (Mehmood et al., *Frontiers in Public Health*, 2021), which proposes CervDetect, a pipeline combining Random Forest feature selection with a shallow neural network to predict cervical cancer biopsy outcomes from the UCI risk-factor dataset. We reimplemented the preprocessing, feature-selection and classification stages in open-source software from the published description. The published confusion matrix reproduces exactly: ours is identical, cell for cell, across all 858 observations, with 803 true negatives, 55 false negatives, and zero true or false positives. The reproduced classifier therefore assigns every observation to the negative class, and its 93.6% majority-class baseline exactly. The all-negative collapse is the modal outcome across random initialisations, though qualitatively different collapse modes are also observed. We further find that the original's Table 4 reports a true-positive rate of 100% on its own published confusion matrix. We conclude that the reported accuracy is reproducible but uninformative, and that the study's detection claims are not supported by its own results.

**A replication of** Mavra Mehmood, Muhammad Rizwan, Michal Gregus ml, Sidra Abbas. Machine Learning Assisted Cervical Cancer Detection. *Frontiers in Public Health*, 9, 788376..

## 1 Introduction

Cervical cancer remains a major cause of mortality among women worldwide, and early identification of risk factors is central to reducing disease burden. Clinical and demographic variables carry useful information about individual risk, and machine learning methods have accordingly been explored as tools for modelling the complex relationships among them. Such datasets are, however, strongly imbalanced: confirmed positive cases are rare relative to negatives, so a classifier that never predicts the positive class can still report high accuracy. Evaluation practice therefore matters as much as model design.

Mehmood et al. [1] propose CervDetect, a machine-learning pipeline for predicting cervical cancer biopsy outcomes using the UCI cervical cancer risk-factors dataset, comprising 858 patient records and 36 attributes. The pipeline proceeds in three stages: treatment of missing values using rules informed by Pearson correlations and domain assumptions, feature selection through Random Forest importance, and classification using a shallow neural network. The authors report an overall accuracy of 93.6% and a mean squared error of 0.07111, while also reporting a false-negative rate of 100%. Despite the tension between the high overall accuracy and the complete absence of positive-case detection, the reported accuracy has subsequently been cited as evidence of the effectiveness of the approach.

These reported results warrant closer examination of the underlying computational procedure. Reproducing a machine-learning study requires more than recovering a headline accuracy: the preprocessing, feature selection, model configuration and evaluation

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The authors have declared that no competing interests exist.

Code is available at <https://github.com/Prakhar-Sahu26/cervdetect-rescience..>

Data is available at <https://archive.ics.uci.edu/dataset/383/cervical+cancer+risk+factors>.

procedure must each be specified in sufficient detail to be reconstructed independently. Several aspects of the CervDetect description do not meet this standard. Two of the stated imputation rules depend on one another for their inputs, one variable is imputed without any rule being given, and the number of selected features differs between the text and the accompanying table. Reported metrics also differ across sections of the paper.

We therefore undertake an independent reconstruction of the CervDetect pipeline, following the preprocessing, feature-selection and classification procedures as described. The reproduction is verified at multiple stages by comparing intermediate outputs—missing-value handling, feature-importance rankings, selected input features and model predictions—against the corresponding results reported in the original study. Where the published description permits more than one reasonable implementation, we make each choice explicit and evaluate its effect through targeted sensitivity analyses.

We find that the published confusion matrix reproduces exactly: ours is identical, cell for cell, across all 858 observations. The reproduced classifier assigns every observation to the negative class, and its accuracy is equal to the majority-class baseline. The all-negative collapse is the modal outcome across random initialisations, though qualitatively different collapse modes are also observed. The result holds across every interpretation of the ambiguous implementation details that we tested.

## 2 Reproduction Methodology

### 2.1 Data

The original study uses the Cervical Cancer (Risk Factors) dataset [2] from the UCI Machine Learning Repository, collected at Hospital Universitario de Caracas, Venezuela. We obtained the file `risk_factors_cervical_cancer.csv` directly from the repository and used it without modification.

The dataset comprises 858 patient records described by 36 attributes. These span demographic variables (age), sexual and reproductive history (number of partners, age at first intercourse, number of pregnancies), smoking history (status, duration, intensity), contraceptive use (hormonal contraceptives and intrauterine devices, each with a corresponding duration), sexually transmitted disease history (a set of individual disease indicators together with summary counts and diagnosis timings), and four diagnostic outcomes.

The four outcome variables are Hinselmann, Schiller, Citology and Biopsy. Each records the result of a distinct diagnostic procedure applied to the same underlying condition. Following the original study, Biopsy is taken as the prediction target, where 1 denotes a positive biopsy result and 0 a negative one. The remaining three outcomes are excluded from the predictor set, as retaining them would constitute target leakage: they represent alternative measurements of the condition being predicted rather than antecedent risk factors.

The original text does not state this exclusion explicitly, but it is evident from Figure 2, which displays the correlation structure of the input variables after missing-value handling. That figure contains exactly 32 variables, and none of them is a diagnostic outcome. This matches the 32 features reported in Section 3 and accounts precisely for the difference from the 36 columns in the raw file. We therefore treat the exclusion as part of the original pipeline rather than as a correction introduced by this reproduction, and follow it throughout.

The target is strongly imbalanced: of the 858 records, 803 are biopsy-negative and 55 are biopsy-positive, a positive prevalence of 6.41%. A classifier that assigns every observation to the negative class therefore attains an accuracy of 93.59%. This value is important context for the results that follow, and we report it alongside every accuracy figure in Section 3.

The reported sample size is inconsistent across the original paper. Section 3 states that the dataset contains 858 instances, while Section 4.1 notes that removing records with missing values would reduce the dataset from 858 to 737. Section 5, however, refers to 859 records. We therefore use 858 observations, consistent with the preprocessing procedure described in Section 4.1, which explicitly states an intention to preserve the number of input vectors rather than reduce them. This choice is further supported by the reported training, validation and test set sizes of 600, 129 and 129 observations respectively, which sum to 858 and are consistent with the stated 70/15/15 split.

## 2.2 Missing-Value Treatment

The raw file encodes missing entries as ?, which we read as NaN to force numeric dtypes; string-based replacement preserves an object dtype and causes silent comparison failures in later stages. Missingness is substantial and highly uneven, ranging from 7 entries in one column to 787 in the two diagnosis-timing columns. All per-column counts match those reported in Table 1 of the original study exactly, confirming that both studies operate on the same file.

The original describes nine imputation rules. Two properties of this specification required decisions on our part: the rules are not stated in an executable order, and several depend on quantities that are themselves subject to imputation. We resolved these as described below, and record every choice in Table 1.

**Execution order.** Rule 4 derives the *STDs* indicator from two individual disease columns, while Rule 5 imputes those same columns by reference to *STDs*. The three columns share an identical 105-row missingness pattern, so each rule requires as input precisely the values the other produces. No execution order satisfies both. We ran Rule 5 first, this being the only order under which Rule 4 has defined inputs, and record the deviation from the stated sequence. Under this order all 105 rows receive *STDs* = 0.

**Reference statistics.** Rules 2 and 3 depend on the means of *Age* and *Num of pregnancies*, the latter of which has 56 missing values imputed under Rule 1. The original does not state whether these means precede or follow imputation. We computed them afterwards, over all 858 rows. This ordering is corroborated by the original’s own reported figures: the claims that 80% of IUD users are older than the mean age and that 70% of non-users have fewer births than the mean reproduce at 79.5% and 69.6% respectively, agreeing to within half a percentage point.

**IUD imputation.** Rule 3 is the least determinate step in the pipeline. The original supplies two conditional proportions of the form  $P(\text{covariate} \mid \text{IUD})$ , but imputation requires  $P(\text{IUD} \mid \text{covariate})$ , and no base rate is given by which to invert them. No assignment rule is stated. We evaluated three candidate readings against the 741 records with observed IUD status, selecting on agreement with those observed values before any model was fitted. A cascade rule—assigning 0 where age is at or below the mean, and applying the pregnancy condition otherwise—agreed with 74.9% of observed values, against 73.3% for a conjunctive reading and 17.7% for a disjunctive one. The cascade rule also gave the highest positive-class recall (0.49) and  $F_1$  (0.31), and yields a post-imputation IUD rate of 11.8% against 11.2% observed. All three readings are carried through to the sensitivity analysis. We note that even the best-performing candidate reproduces only three quarters of observed values, so approximately 25% of the 117 imputed entries are expected to be incorrect under any available reading.

**An unaddressed variable.** No rule in the original addresses *STDs:HIV*. Rule 1 excludes it by its missingness count, and Rules 6–8 name AIDS, Hepatitis B and HPV individually

**Table 1.** Imputation rules, ambiguities and resolutions.

| Rule                 | NaN  | Ambiguity                                                                             | Resolution                                                                  |
|----------------------|------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 1 – median fill      | 7–56 | Reference set and split timing unstated; <i>Smokes</i> is binary but median specified | Column median over all 858 rows; median taken as written                    |
| 2 – HC               | 108  | Correlation rationale not executable; only the final condition is implementable       | Applied stated condition to missing rows only                               |
| 2b – HC (years)      | 108  | Global vs. within-group median unstated                                               | Within-group median; shared missingness makes the reference set unambiguous |
| 3 – IUD              | 117  | Conditionals inverted; no assignment rule given                                       | Cascade rule, selected on agreement with observed values                    |
| 3b – IUD (years)     | 117  | Cites a –1 sentinel absent from the source; mean here vs. median in Rule 2b           | Followed the stated mean                                                    |
| 4 – STDs             | 105  | Inputs unavailable on exactly the rows to be filled; circular with Rule 5             | Rule 5 executed first                                                       |
| 5 – STD indicators   | 105  | Mean and median both specified; column list not enumerated                            | Column median on the eight indicators                                       |
| 6 – AIDS             | 105  | Fill value not stated                                                                 | Zero; column is constant                                                    |
| 7 – Hepatitis B      | 105  | None                                                                                  | Zero, as stated                                                             |
| 8 – HPV              | 105  | None                                                                                  | Zero, as stated                                                             |
| – HIV                | 105  | No rule given                                                                         | Column median                                                               |
| 9 – Diagnosis timing | 787  | Covers only $STDs = 0$                                                                | Stated rule, plus group median for the 8 remaining                          |

but not HIV. The variable nonetheless appears in Figure 2 of the original, so it appears to have been retained through preprocessing by an unreported procedure. We filled it with the column median, consistent with reading it under Rule 5’s reference to STD indicators generally. As the median is 0, this reading and a direct zero-fill produce an identical column.

**Partial rule coverage.** Rule 9 imputes the two diagnosis-timing columns from the premise that a negative STD status implies no prior diagnosis. This covers only the 779 records with  $STDs = 0$ ; the remaining 8 in each column have  $STDs = 1$  and receive no instruction. Filling them with zero would contradict the rule’s own premise. We used the within-group median of observed STD-positive records, retaining zero-fill as a sensitivity variant. The affected records are 0.9% of the sample and contain one biopsy-positive case against 0.5 expected at the base rate.

After all nine rules and the HIV fill, the feature frame contains no missing values and comprises  $858 \times 32$  entries, matching both the variable count in Figure 2 of the original and the 32 features reported in its Section 3.

## 2.3 Feature Selection

The original uses a Random Forest classifier to rank the 32 input variables by impurity-based importance, and selects the highest-ranked subset as input to the neural network. The Random Forest itself is not used for prediction. Because the original does not report RF hyperparameters or a random seed, we used `RandomForestClassifier` with `n_estimators=100` and `random_state=42`, otherwise default settings, fitted to the complete  $858 \times 32$  frame with Biopsy as the target.

Following the pipeline order given in Figure 1 of the original, feature selection precedes the train–test split. The ranking is therefore computed on data that includes the eventual test partition. We retain this ordering for fidelity, but note that this constitutes

information leakage from the eventual test partition and can make subsequent test performance optimistically biased.

Despite the unreported hyperparameters, the resulting ranking, shown in Figure 1, agrees closely with Figure 3 of the original. The five highest-ranked variables—age, duration of hormonal contraceptive use, age at first intercourse, number of sexual partners and number of pregnancies—appear in the same order in both rankings. The ten highest-ranked variables in our ranking coincide with those shown in the top-ten region of Figure 3, differing only in the ordering of closely ranked variables. This should not be confused with Table 3, whose ten selected variables do not correspond to the top ten of the reported ranking. The lowest-ranked variables form the same near-zero tail in both. Across all variables the Spearman rank correlation between our ranking and the original's is  $\rho = 0.975$ . We take this agreement as evidence that the preprocessing stage has been reconstructed faithfully, since the ranking depends on every imputation decision described in Section 2.2.

The size of the selected subset is reported inconsistently. The text states that 12 optimum features were used as the network input, whereas Table 3 of the original lists ten. The two additional variables are not identified anywhere in the paper.

The discrepancy is not resolved by taking the ten highest-ranked variables. Table 3 omits the number of sexual partners, which is ranked fourth in both our ranking and Figure 3, while including the time since last STD diagnosis, which is ranked lower. Table 3 is therefore not a truncation of the reported importance ranking, and the selection procedure that produced it is not fully described by the Random Forest stage alone.

We reconstructed the 12-variable set as the ten listed in Table 3 together with the next two variables by importance, namely the number of sexual partners and the STD count. Because neither reading can be confirmed, both the ten-variable and twelve-variable sets are carried through the sensitivity analysis reported in Section 3, and all results are reported for each.

## 2.4 Model Architecture and Training

**Implementation environment of the original.** Several of the choices described below rest on a hypothesis about the software used in the original study, which we state explicitly so that it can be assessed independently. The original reports a shallow network of 50 hidden neurons trained by scaled conjugate gradient, and includes a training-progress figure of the form produced by MATLAB's Neural Network Toolbox. We therefore treat that toolbox, most plausibly through `patternnet` and `trainscg`, as the working software hypothesis for the original implementation. This matters because the toolbox applies several transformations by default that an author would have no occasion to document. Wherever we adopt such a default, we identify it as an inference drawn from this hypothesis rather than as a stated procedure.

**Architecture.** The network has 12 inputs, a single hidden layer of 50 units and a single output unit, with sigmoid activation at both layers as specified in the original's Algorithm 1. Writing  $\sigma$  for the logistic function,

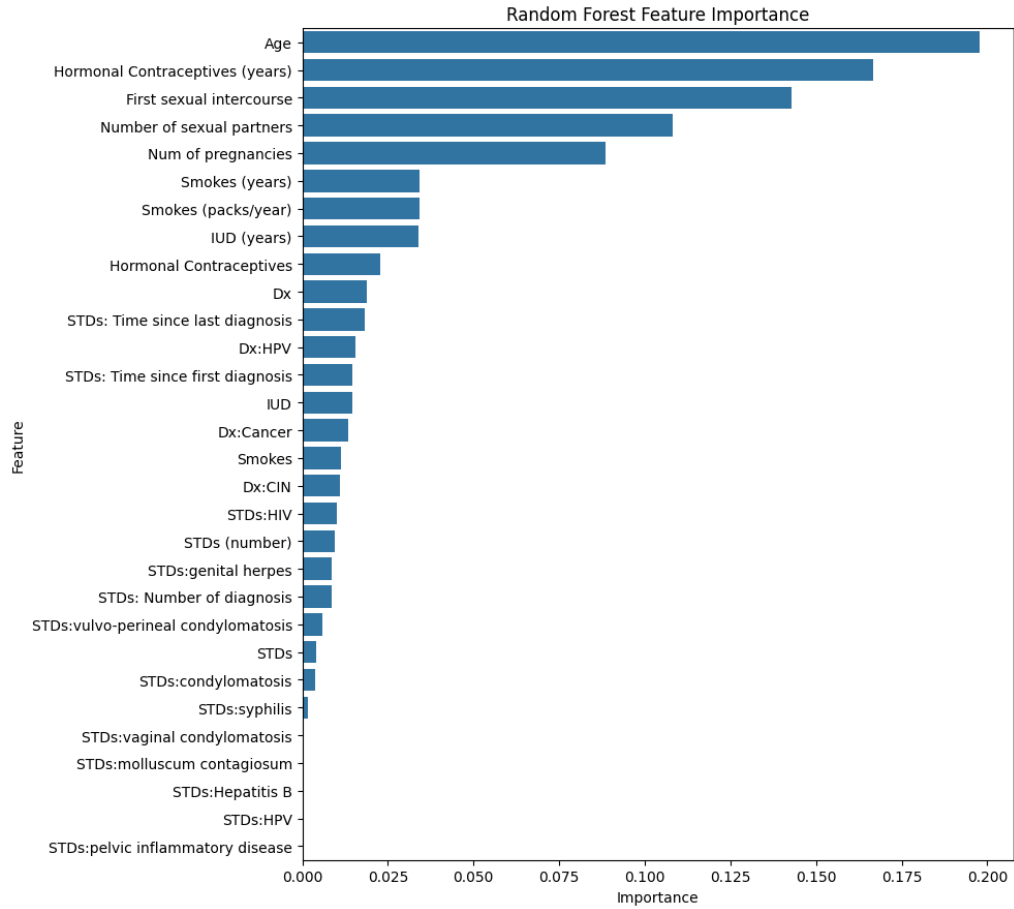
$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad (1)$$

the network computes

$$a^{(1)} = \sigma(W^{(1)}x + b^{(1)}), \quad (2)$$

$$\hat{y} = \sigma(W^{(2)}a^{(1)} + b^{(2)}), \quad (3)$$

with  $W^{(1)} \in \mathbb{R}^{50 \times 12}$  and  $W^{(2)} \in \mathbb{R}^{1 \times 50}$ . Predictions are obtained by thresholding  $\hat{y}$  at 0.5.



**Figure 1.** Random Forest impurity-based feature importance on the reconstructed  $858 \times 32$  frame, ranked in descending order. The Spearman rank correlation with Figure 3 of the original study is  $\rho = 0.975$ .

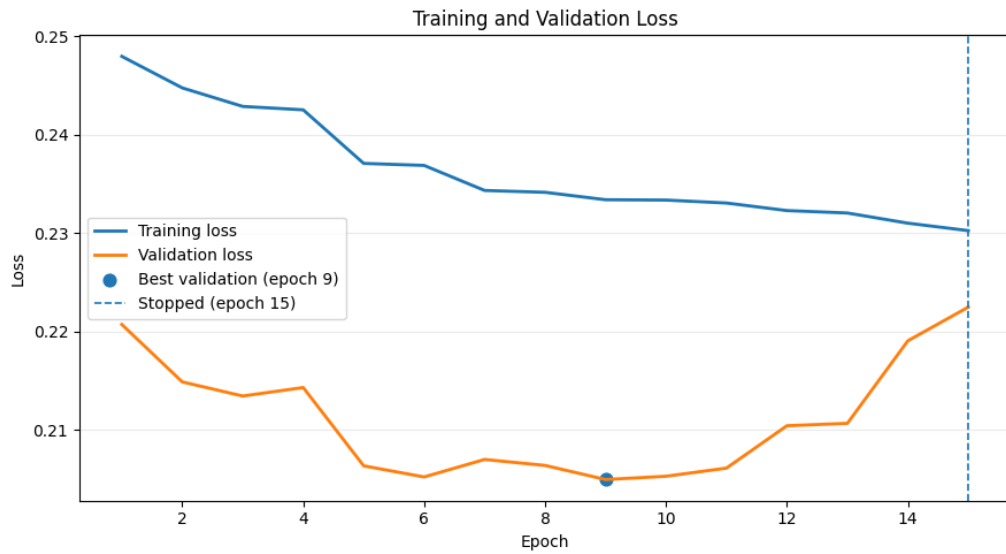
The original’s description of the output layer is not consistent across its equations and figures. We therefore also evaluate a variant with hyperbolic-tangent hidden activation and a two-unit softmax output, corresponding to the toolbox defaults.

**Initialisation.** Algorithm 1 specifies that weights are drawn from  $N(0, 1)$  and bias nodes set to 1.0, and we follow this exactly. A Nguyen–Widrow-style initialisation, corresponding to the toolbox’s `initnw` alternative, is evaluated as a variant. This choice is not incidental: because the derivative of the logistic function,

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)), \quad (4)$$

attains a maximum of 0.25 at  $z = 0$  and decays towards zero as  $|z|$  grows, the gradient available to the first layer is governed by the magnitude of the pre-activations  $W^{(1)}x + b^{(1)}$ . Unit-variance weights applied to inputs on their natural scales place those pre-activations far from the origin, as we quantify below.

**Input normalisation.** No scaling or normalisation step is described anywhere in the original. We nonetheless apply min–max scaling of each input to  $[-1, 1]$ , corresponding to the toolbox’s `mapminmax` default, which is applied inside the training routine and would not ordinarily be reported by an author. The scaler is fitted over all 858 rows,



**Figure 2.** Training and validation loss for the primary configuration (twelve features, sigmoid activation, cross-entropy loss, seed 42). Validation loss attains its minimum at epoch 9 and training terminates at epoch 15 after six consecutive increases. The corresponding figure in the original study shows a best validation epoch of 3 and termination at epoch 9—the same interval of six epochs. Note the vertical scale: training loss falls only from 0.248 to 0.230 across the full run, consistent with the attenuated gradients described in Section 2.4.

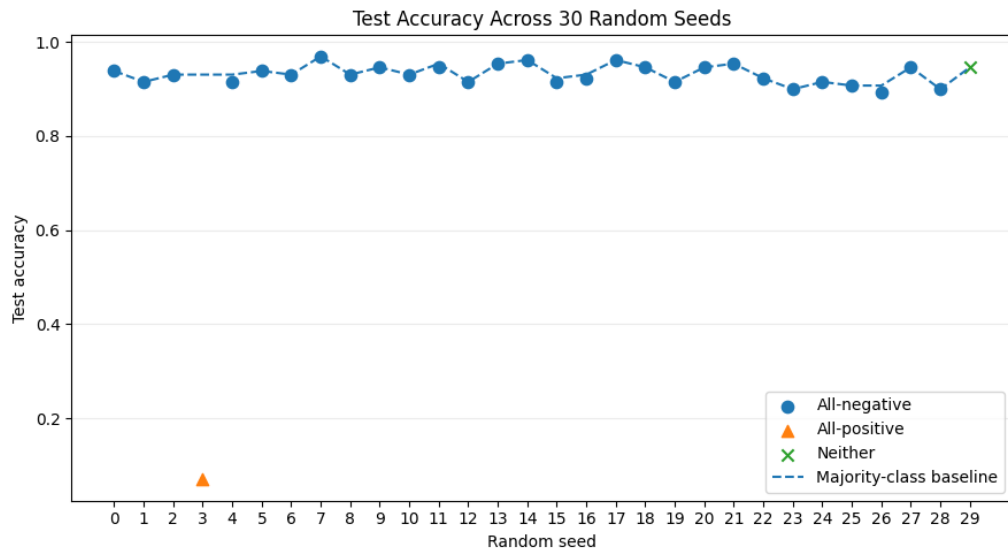
matching the toolbox behaviour of normalising before its internal partition; fitting on the training partition alone is retained as a sensitivity variant.

Omitting this step has a measurable effect on the optimisation regime. The input variables differ by more than an order of magnitude in scale, from binary indicators to ages ranging from 13 to 84, and the mean Euclidean norm of an input row is 32.9. Under the initialisation of Algorithm 1 this yields hidden-layer pre-activations with a standard deviation of 32.2, and 90.0% of hidden-layer activations fall below 0.01 or above 0.99 before any gradient step is taken. By Equation (4) the mean sigmoid derivative in this regime is 0.0102, against a maximum of 0.25. Holding the initial weights fixed and rescaling the inputs alone, the pre-activation standard deviation falls to 2.9, the saturated fraction to 17.3%, and the mean derivative rises to 0.0968—a factor of approximately 9.5. Without normalisation the network begins training with approximately 90% of hidden units in the saturated region and a mean sigmoid derivative approximately 9.5-fold smaller than with scaled inputs; this is consistent with the optimisation difficulty we observe in its absence (Section 3.6).

**Loss function.** The original specifies squared error in Algorithm 1 and reports minimised cross-entropy in its results section. We use binary cross-entropy in the primary configuration and evaluate mean squared error as a variant.

**Training algorithm.** We implemented the scaled conjugate gradient algorithm of Møller directly, including the finite-difference approximation to the Hessian-vector product and the Levenberg–Marquardt-style scaling parameter, rather than substituting a general-purpose conjugate gradient routine; the latter performs a line search at each step and is a different algorithm.

**Stopping criterion.** The training duration is reported inconsistently: the text states one epoch, while the accompanying training figure shows nine. Rather than fixing the



**Figure 3.** Test accuracy across 30 random seeds, with points marked by the behaviour of the resulting classifier. Twenty-eight runs assign every observation to the negative class and attain accuracy equal to the majority-class baseline of the corresponding split. One run (seed 3) assigns every observation to the positive class, giving a true-positive rate and a false-positive rate of 100% at an accuracy equal to the positive prevalence of its split. One run falls into neither category.

iteration count at either value, we adopted the toolbox default of terminating after six consecutive increases in validation loss, subject to a cap of 1000 epochs, and allowed training to halt of its own accord. The stopping epoch is thereby an observation rather than an assumption; Figure 2 shows the resulting trajectory.

**Data partition.** The original reports a 70/15/15 split into training, validation and test sets of 600, 129 and 129 observations. Neither the random seed nor whether the split was stratified is reported. We used an unstratified random split as an analogue of the toolbox’s default random division, with seed 42 in the primary run. This seed is our own choice and carries no significance; the entire procedure is repeated across 30 seeds (Figure 3) to establish that the behaviour reported below does not depend on it. In the primary run the three partitions contain 40, 7 and 8 positive cases respectively.

**Procedures not applied.** The original describes no class weighting, resampling, synthetic oversampling or threshold adjustment, and we applied none in the reproduced classifier. A Youden-optimal threshold is computed in Section 3 as a diagnostic only and forms no part of the reproduction. We state this explicitly because each of these procedures would materially alter the behaviour reported below, and their absence is a property of the original design rather than of our reconstruction.

**Configurations evaluated.** The ambiguities identified above admit more than one defensible reading. Rather than selecting among them, we evaluate the full factorial combination of feature set (ten or twelve variables), architecture and loss (sigmoid with cross-entropy, sigmoid with squared error, or tanh-softmax with cross-entropy) and initialisation ( $N(0, 1)$  or Nguyen-Widrow), giving twelve configurations, each repeated across ten seeds. Results for all twelve are reported in Section 3.

## 2.5 Evaluation Protocol

**Metric definitions.** We adopt the metric definitions given in the original study. Writing TP, TN, FP and FN for the counts of true positives, true negatives, false positives and false negatives,

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (5)$$

$$\text{TPR} = \frac{TP}{TP + FN}, \quad (6)$$

$$\text{FPR} = \frac{FP}{FP + TN}, \quad (7)$$

$$\text{FNR} = \frac{FN}{FN + TP}. \quad (8)$$

We state these explicitly because the relations they impose are used in Section 3 to evaluate the internal consistency of the originally reported figures. In particular, Equations (6) and (8) give  $\text{TPR} + \text{FNR} = 1$  whenever the positive class is non-empty, and for a confusion matrix with both classes represented these quantities are uniquely determined. Precision,  $TP/(TP + FP)$ , and the  $F_1$  score are reported additionally; neither appears in the original.

**The majority-class baseline.** Accuracy is not interpretable in isolation on an imbalanced dataset, since it can be made large by ignoring the minority class entirely. For each partition we therefore also report the accuracy of the trivial classifier that assigns every observation to the more frequent class, which is the larger of the two class proportions in that partition. On the full dataset this quantity is 93.59%. Every accuracy figure reported in Section 3 is accompanied by the corresponding baseline and by the difference between the two, which we refer to as the lift over baseline.

**Threshold-free metrics.** The metrics above are computed from thresholded predictions and therefore describe the classifier only at the particular threshold applied. To characterise the model’s ranking behaviour independently of that choice, we additionally report the area under the receiver operating characteristic curve (ROC-AUC) and the average precision (AP), computed from the continuous network outputs.

The two require different reference points. ROC-AUC takes the value 0.5 in expectation for a classifier that ranks at random, irrespective of class balance. Average precision does not share this property: for a random ranking the expected average precision is the positive prevalence of the sample, here 0.064. An average precision figure is therefore reported alongside the prevalence of the partition it was computed on, since the same numerical value carries different meaning at different prevalences.

**Metrics reported by the original.** The original reports accuracy, mean squared error, true-positive rate, false-positive rate and false-negative rate, together with a confusion matrix and a set of ROC curves. It does not report the area under those curves, average precision, precision,  $F_1$ , any majority-class baseline, or any measure of variability across repeated runs; results are given for a single run whose random seed is not stated. Our protocol reports the classification and threshold-free quantities given in the original, except for the reported mean squared error, which we treat separately because its calculation cannot be uniquely reconstructed from the published information. We add the baseline comparison and the threshold-free metrics described above.

## 2.6 Software and Environment

The reproduction was implemented in Python. The original is, under the software hypothesis stated in Section 2.4, treated as a MATLAB implementation; reimplementation in open-source software is therefore itself part of this work, and any residual difference in numerical behaviour between the two environments cannot be excluded as a source of discrepancy.

Data handling and preprocessing use `pandas` and `NumPy`. The Random Forest of Section 2.3 and all evaluation metrics use `scikit-learn`. The network, the scaled conjugate gradient routine and the initialisation schemes were implemented directly using `PyTorch` tensors and automatic differentiation; no built-in optimiser was used. Figures were produced with `Matplotlib` and `seaborn`. Package versions are pinned in `requirements.txt` in the repository.

The stochastic components are seeded explicitly. The primary Random Forest is fitted with a fixed seed, while the data partition and the neural-network initialisation use the corresponding run seed. The primary run uses seed 42 throughout; the repetition analyses use seeds 0 to 29 and 0 to 9 as described in Section 2.4. With fixed seeds and environment, the implemented pipeline is intended to be reproducible, and re-execution reproduces the figures and tables reported from the notebook.

The complete code, including the preprocessing rules, the sensitivity variants and the scripts that generate each figure, is available in the repository cited above. The dataset is not redistributed and is obtained directly from the UCI repository.

## 3 Results

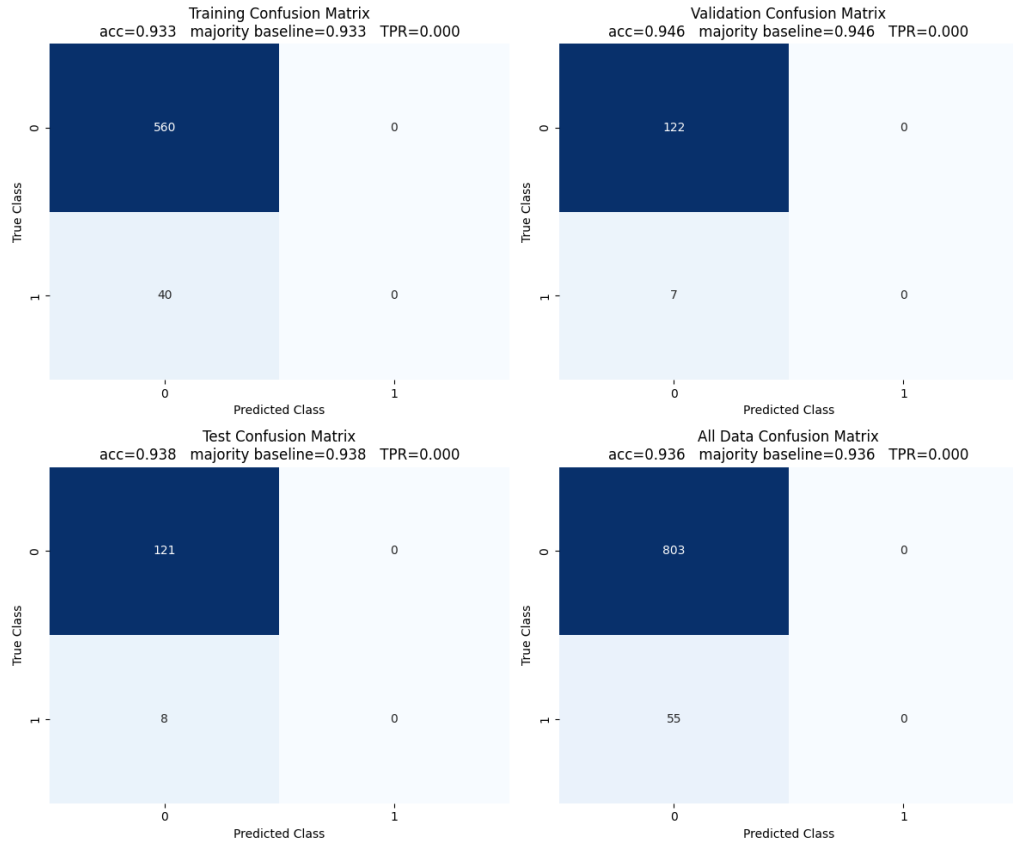
### 3.1 Reproduction of the Reported Confusion Matrix

The confusion matrices produced by the primary configuration are shown in Figure 4. Across all three partitions the reproduced classifier predicts the negative class for every observation: the training partition yields 560 true negatives and 40 false negatives, the validation partition 122 and 7, and the test partition 121 and 8. No observation in any partition is assigned to the positive class, so the false-positive and true-positive counts are zero throughout.

Pooling the three partitions gives 803 true negatives, 55 false negatives, and no true or false positives across all 858 observations. This is identical, cell for cell, to the confusion matrix reported in Figure 6 of the original study. The per-partition counts differ slightly from those reported there—the original gives 559 and 41 for training, and 123 and 6 for validation, against our 560/40 and 122/7—but these differences reflect a different random allocation of the same records between partitions, not a difference in classifier behaviour. The test partition matches exactly at 121 and 8, and the pooled totals are identical.

The reproduction therefore recovers the published classification behaviour rather than approximating it. Both the published model and its reconstruction assign every one of the 858 observations to the negative class, and neither identifies any of the 55 biopsy-positive cases.

Training terminated at epoch 15, with the lowest validation loss attained at epoch 9 (Figure 2). The original's training figure shows a best validation epoch of 3 and termination at epoch 9. The interval between the two is six epochs in both cases, consistent with the stopping rule adopted in Section 2.4. This also bears on the discrepancy noted earlier between the original's text, which states one epoch, and its training figure, which shows nine. The six-epoch interval is consistent with the stopping rule adopted here, but does not establish how epochs were counted in the original implementation.



**Figure 4.** Confusion matrices for the primary configuration (twelve features, sigmoid activation, cross-entropy loss, seed 42), by partition and pooled across all 858 observations. In every panel the accuracy equals the majority-class baseline of that partition and the true-positive rate is zero. The pooled matrix is identical, cell for cell, to Figure 6 of the original study.

### 3.2 Accuracy Relative to the Majority-Class Baseline

Table 2 reports the accuracy of the reproduced classifier alongside the majority-class baseline for each partition, as defined in Section 2.5.

In every partition the two quantities are equal and the lift over baseline is zero. This follows directly from the confusion matrices of Section 3.1: a classifier that assigns every observation to the negative class attains exactly the accuracy of the trivial classifier that does the same, since the two make identical predictions on every record. The pooled accuracy of 93.59% is the negative-class prevalence of the dataset.

The original reports an accuracy of 93.6% and does not report a baseline for comparison. That figure agrees with both the accuracy of the reproduced classifier and the accuracy attainable without reference to any input variable.

### 3.3 Internal Consistency of the Originally Reported Metrics

The original reports classification metrics in three places: the abstract, the results table (Table 4), and the confusion matrix (Figure 6). The values implied by each are collected in Table 3.

The confusion matrix determines the remaining quantities uniquely. With  $TP = 0$ ,  $FP = 0$ ,  $FN = 55$  and  $TN = 803$ , Equations (6)–(8) give  $TPR = 0/55 = 0$ ,  $FPR = 0/803 = 0$  and  $FNR = 55/55 = 1$ , with an accuracy of  $803/858 = 0.9359$ . The reported accuracy of 93.6% agrees with this to the precision given, as does the false-negative rate

**Table 2.** Accuracy of the reproduced classifier against the majority-class baseline, by partition. Lift is the difference between the two.

| Partition  | $n$ | Pos. | Accuracy | Baseline | Lift   |
|------------|-----|------|----------|----------|--------|
| Training   | 600 | 40   | 0.9333   | 0.9333   | 0.0000 |
| Validation | 129 | 7    | 0.9457   | 0.9457   | 0.0000 |
| Test       | 129 | 8    | 0.9380   | 0.9380   | 0.0000 |
| All data   | 858 | 55   | 0.9359   | 0.9359   | 0.0000 |

**Table 3.** Classification metrics as reported in three locations of the original study, and as implied by its published confusion matrix. Dashes indicate quantities not reported in that location.

| Source             | Acc.  | TPR  | FPR  | FNR  |
|--------------------|-------|------|------|------|
| Abstract           | 93.6% | —    | 6.4% | 100% |
| Table 4            | 93.6% | 100% | 100% | —    |
| Figure 6 (implied) | 93.6% | 0%   | 0%   | 100% |
| This reproduction  | 93.6% | 0%   | 0%   | 100% |

of 100% reported in the abstract.

The remaining reported values do not follow. Table 4 gives a true-positive rate of 100%, whereas the confusion matrix requires zero; the two are related by  $TPR + FNR = 1$ , so a true-positive rate of 100% and the abstract's false-negative rate of 100% cannot both hold. Table 4 further gives a false-positive rate of 100%, whereas the confusion matrix requires zero and the abstract reports 6.4% for the same quantity.

The value 6.4% is consistent with a different quantity. The proportion of misclassified observations is  $(FP + FN)/n = 55/858 = 0.0641$ , which agrees with the reported figure to the precision given, whereas Equation (7) applied to the published confusion matrix gives zero.

No assignment of values to  $TP$ ,  $TN$ ,  $FP$  and  $FN$  satisfies the metrics reported in Table 4 and those reported in the abstract simultaneously. If the three sources are intended to describe the same run at the same operating point, their reported values are mutually incompatible, and at least one must refer to a different result, contain an error, or arise from a different calculation.

### 3.4 Stability Across Random Initialisations

The primary run reported above uses seed 42. To establish whether its behaviour is characteristic of the pipeline or an artefact of that choice, the entire procedure—partition, initialisation and training—was repeated across 30 seeds. Test accuracy for each run is shown in Figure 3, and the distribution of test metrics is summarised in Table 4.

Twenty-eight of the 30 runs assign every observation to the negative class, giving a true-positive rate of zero. The median true-positive rate across the 30 runs is 0.000, with an interquartile range of [0.000, 0.000]; the median  $F_1$  score is likewise 0.000. Median test accuracy is 0.930, with an interquartile range of [0.915, 0.946], tracking the majority-class baseline of each partition as the number of positive cases allocated to the test set varies between 4 and 13.

The two remaining runs depart from this pattern in opposite directions. One falls into neither category, recovering a small number of positive cases. The other, seed 3, assigns every observation to the *positive* class: on a test partition containing 120 negative and 9 positive records it returns  $TP = 9$ ,  $FP = 120$ ,  $TN = 0$  and  $FN = 0$ , giving a true-positive rate of 100%, a false-positive rate of 100% and an accuracy of  $9/129 = 0.070$ , equal to the positive prevalence of that partition. Training in this run terminated at

**Table 4.** Distribution of test-set metrics across 30 random seeds.

| Metric                        | Median | IQR            | Min   | Max   |
|-------------------------------|--------|----------------|-------|-------|
| Accuracy                      | 0.930  | [0.915, 0.946] | 0.070 | 0.969 |
| TPR                           | 0.000  | [0.000, 0.000] | 0.000 | 1.000 |
| FNR                           | 1.000  | [1.000, 1.000] | 0.000 | 1.000 |
| $F_1$                         | 0.000  | [0.000, 0.000] | 0.000 | 0.222 |
| ROC-AUC                       | 0.516  | [0.450, 0.577] | 0.273 | 0.739 |
| Average precision             | 0.091  | [0.075, 0.123] | 0.039 | 0.176 |
| Runs with $TP = 0$ : 28 of 30 |        |                |       |       |

epoch 1 with the lowest validation loss recorded at epoch 0, indicating that no trained epoch improved on the initial validation loss.

The all-negative outcome is therefore the typical behaviour of the pipeline rather than a property of the particular seed used in the primary run. The distribution of test metrics is also strongly affected by the small number of positive cases available: across the 30 partitions the test set contains between 4 and 13 positive records, so the recovery of a single additional positive case changes the true-positive rate by between 8 and 25 percentage points, and any recall figure derived from a single run carries substantial variance.

### 3.5 Sensitivity to Ambiguous Implementation Choices

The results reported use a single reconstruction of the original pipeline. Three of the ambiguities identified in Section 2 admit more than one defensible reading: the size of the selected feature set, which the original states as twelve but tabulates as ten; the architecture and loss function, specified as sigmoid activation with squared error in Algorithm 1 but reported as cross-entropy in the results section, with the toolbox defaults implying a hyperbolic-tangent hidden layer and softmax output; and the initialisation scheme, given as  $N(0, 1)$  in Algorithm 1 against the toolbox’s Nguyen–Widrow alternative. These give twelve reconstructions, each of which follows the published description as closely as the description permits.

All twelve were evaluated over ten seeds. Median test metrics are reported in Table 5.

Every configuration returns a median true-positive rate of 0.000. No combination of feature set, architecture, loss function and initialisation recovers any positive case at the median, and median accuracy lies between 0.930 and 0.934 throughout, matching the majority-class baseline of the test partition in each case.

The configurations are not otherwise equivalent. Median ROC-AUC ranges from 0.425 to 0.595, and the median stopping epoch from 9 to 18.5, so the twelve reconstructions produce measurably different models. On the twelve-feature set, Nguyen–Widrow initialisation yields a higher median ROC-AUC than  $N(0, 1)$  in all three architecture and loss pairings, by between 0.088 and 0.155; on the ten-feature set the difference is smaller and not consistent in sign. Three configurations return a median ROC-AUC below 0.5. We do not identify any of these configurations as the correct reconstruction, since the published description does not determine which was used.

The variation in ranking behaviour therefore does not translate into any variation in classification outcome. The all-negative result reported in Sections 3.1 and 3.4 does not depend on how the ambiguities in the original description are resolved, and the inconsistencies identified in the original text need not be settled in order to reproduce its reported behaviour.

**Table 5.** Median test metrics across ten seeds for each of the twelve reconstructions. V1: sigmoid hidden and output with binary cross-entropy. V1b: as V1 with mean squared error. V2: hyperbolic-tangent hidden layer with two-unit softmax output and cross-entropy.

| Features                               | Init.     | Acc.  | TPR   | AUC   | Epoch |
|----------------------------------------|-----------|-------|-------|-------|-------|
| <i>V1: sigmoid, cross-entropy</i>      |           |       |       |       |       |
| Twelve                                 | $N(0, 1)$ | 0.930 | 0.000 | 0.455 | 16.5  |
| Twelve                                 | N-W       | 0.930 | 0.000 | 0.595 | 12.5  |
| Ten                                    | $N(0, 1)$ | 0.930 | 0.000 | 0.474 | 14.0  |
| Ten                                    | N-W       | 0.930 | 0.000 | 0.480 | 11.0  |
| <i>V1b: sigmoid, squared error</i>     |           |       |       |       |       |
| Twelve                                 | $N(0, 1)$ | 0.930 | 0.000 | 0.425 | 15.0  |
| Twelve                                 | N-W       | 0.930 | 0.000 | 0.580 | 14.0  |
| Ten                                    | $N(0, 1)$ | 0.934 | 0.000 | 0.503 | 14.5  |
| Ten                                    | N-W       | 0.930 | 0.000 | 0.503 | 11.0  |
| <i>V2: tanh-softmax, cross-entropy</i> |           |       |       |       |       |
| Twelve                                 | $N(0, 1)$ | 0.930 | 0.000 | 0.495 | 14.0  |
| Twelve                                 | N-W       | 0.930 | 0.000 | 0.583 | 10.0  |
| Ten                                    | $N(0, 1)$ | 0.930 | 0.000 | 0.489 | 18.5  |
| Ten                                    | N-W       | 0.930 | 0.000 | 0.444 | 9.0   |

### 3.6 Ranking Behaviour and the Effect of Input Normalisation

The results above describe the classifier at the 0.5 threshold. This section examines the continuous outputs from which those predictions are derived, and the effect of the normalisation step introduced in Section 2.4.

**Effect of normalisation.** Without normalisation the network’s outputs are almost undifferentiated between classes: the difference in mean predicted probability between biopsy-positive and biopsy-negative records is +0.0228, and the difference in medians is +0.0011. Test ROC-AUC for that run is 0.507. With normalisation the median gap increases to +0.0084 and test ROC-AUC to 0.583. Average precision was not recorded for the unnormalised run and is therefore not compared.

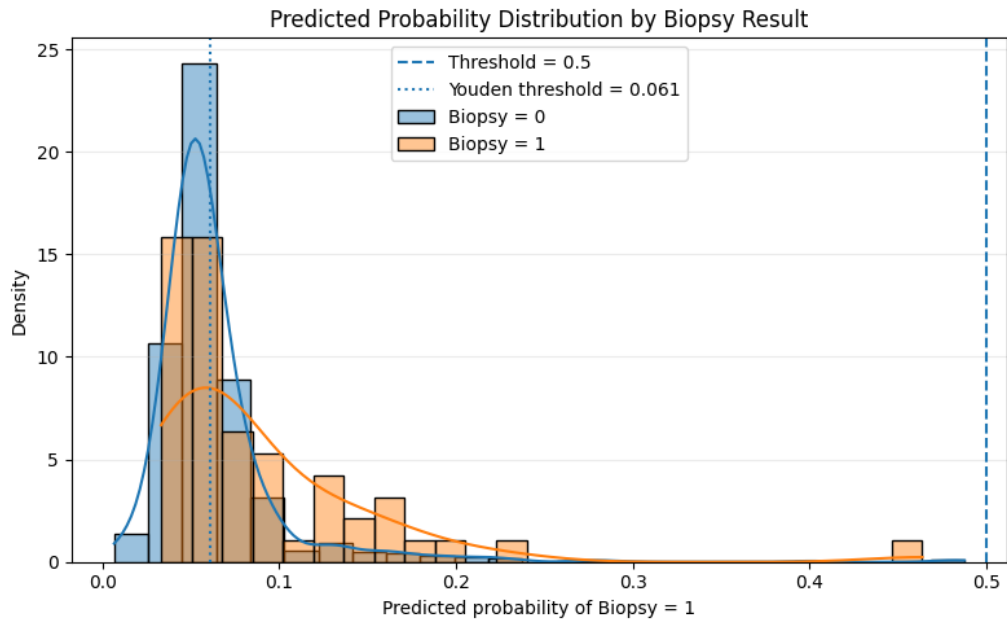
The normalised model also matches the original more closely, not less. The unnormalised run recovers one positive case on the test partition, giving a test confusion matrix of  $TN = 121$ ,  $FP = 0$ ,  $FN = 7$ ,  $TP = 1$ ; the normalised run recovers none, reproducing the published matrix exactly (Section 3.1). An inferred preprocessing step that moves the reconstruction towards the published result, rather than away from it, is evidence that a comparable step was present in the original pipeline.

**Threshold-free performance.** ROC-AUC and average precision for the primary run are given in Table 6. Pooled across all 858 observations, ROC-AUC is 0.622 and average precision 0.128 against a prevalence of 0.064. Both exceed their reference values, so the model’s outputs carry some information about the target: ranking performance is above the random-ranking reference, and average precision is roughly twice the prevalence baseline.

That information does not survive thresholding. The distribution of predicted probabilities by class is shown in Figure 5. Positive records receive slightly higher probabilities on average—mean 0.0883 against 0.0634, median 0.0630 against 0.0546—but the two distributions show substantial overlap. The highest probability assigned to any of the 858 observations is 0.4877, and it belongs to a biopsy-negative record; the highest assigned to any positive record is 0.4638. Both fall below 0.5, so no observation can be classified positive at that threshold regardless of the ranking information the outputs contain.

**Table 6.** Threshold-free metrics for the primary run. Average precision is compared against the positive prevalence of each partition, which is its expected value under a random ranking.

| Partition  | ROC-AUC | AP    | Prevalence |
|------------|---------|-------|------------|
| Training   | 0.638   | 0.154 | 0.067      |
| Validation | 0.562   | 0.146 | 0.054      |
| Test       | 0.583   | 0.110 | 0.062      |
| All data   | 0.622   | 0.128 | 0.064      |

**Figure 5.** Distribution of predicted probabilities by biopsy outcome, pooled across all 858 observations. The dashed line marks the 0.5 classification threshold. No observation of either class receives a probability above 0.4877.

**Threshold sensitivity.** The threshold maximising Youden’s  $J$  on the test partition is 0.061, close to the positive prevalence of the dataset. At that threshold the classifier attains a true-positive rate of 0.625 and a false-positive rate of 0.331, recovering five of the eight positive cases in the test partition. This is reported as a diagnostic only; the reproduced classifier uses the 0.5 threshold throughout, as described in Section 2.4.

## 4 Discussion

### 4.1 What Reproduces and What Does Not

The reconstructed pipeline reproduces the published confusion matrix, but not all of the reported metrics, and these two outcomes are worth separating.

The confusion matrix reproduces exactly. Both the published matrix and ours contain 803 true negatives, 55 false negatives, and no true or false positives across all 858 observations. This agreement holds despite a reimplement in different software, an unreported random seed, and several points at which the published description had to be interpreted (Section 3.5). We therefore regard the published thresholded prediction behaviour as successfully reproduced. Identical predictions do not establish that the two implementations are identical in every internal respect, but they do establish that

the reported classification result is recoverable from the published description.

What that classifier does is assign every observation to the negative class. Its accuracy of  $803/858 = 93.59\%$  is exactly the proportion of negative cases in the dataset: the same accuracy is obtained by a classifier that ignores all input variables and predicts the negative class for every observation. The reproduction therefore recovers the original accuracy result while establishing that the reported thresholded classifier provides no positive-case detection beyond the majority-class baseline. Reproducibility and informativeness are independent properties, and this study is an instance in which the first holds and the second does not.

The reported metrics do not reproduce. The values in the original's Table 4 cannot be derived from its own published confusion matrix, and are inconsistent with those reported in its abstract (Section 3.3). Our reproduction agrees with the confusion matrix and with the abstract's false-negative rate of 100%, and disagrees with Table 4. If Table 4 and Figure 6 are intended to describe the same run at the same operating point, their reported values are mutually incompatible; this is a discrepancy internal to the original rather than a disagreement between the original and this reproduction. Whether they may instead describe different runs is considered in Section 4.2.

A qualification is needed on what the network has learned. The all-negative outcome invites the conclusion that it learned nothing, and this overstates the case. Pooled ROC-AUC is 0.622 and average precision 0.128 against a prevalence of 0.064, so the continuous outputs exhibit weak discrimination above the random-ranking reference. They show weak ranking ability rather than the behaviour expected from a random ranking. They are nevertheless far too weakly separated, at the reported operating point, to support a useful diagnostic classifier: the two probability distributions show substantial overlap, and no observation of either class receives a probability above 0.4877 (Section 3.6). The failure therefore does not imply an absence of all signal; the continuous outputs exhibit weak ranking ability, but under the reported 0.5 threshold that signal does not produce any positive prediction.

The appropriate interpretation is therefore not that the network learned nothing, but that its weak continuous ranking signal does not translate into a useful classifier at the reported threshold. The observed accuracy primarily reflects the class distribution, whereas positive-class detection performance is zero. The distinction between discrimination and thresholded classification matters for interpreting what, if anything, can be salvaged from the approach, which we examine in Section 4.4.

## 4.2 On the Reported Metrics

Section 3.3 established that the metrics reported in the original's Table 4 cannot be derived from its published confusion matrix. This section considers what might account for that.

The reported values are not arbitrary. A true-positive rate of 100% together with a false-positive rate of 100% is the exact signature of a classifier that assigns every observation to the positive class: such a classifier recovers every positive case, giving  $TPR = TP/(TP + FN) = 1$ , while misclassifying every negative case, giving  $FPR = FP/(FP + TN) = 1$ . The pair of values reported in Table 4 is therefore not an arbitrary error but a coherent description of a specific degenerate outcome—one that is the exact class-prediction complement of the all-negative outcome shown in the same paper's Figure 6.

That outcome is attainable under our reconstructed implementation of this architecture. Among the 30 initialisations reported in Section 3.4, one produced precisely this behaviour: on a test partition of 120 negative and 9 positive records it returned  $TN = 0$ ,  $FP = 120$ ,  $FN = 0$  and  $TP = 9$ , giving a true-positive rate and a false-positive rate of 100% at an accuracy of 0.070. Under our reconstructed implementation the model can therefore collapse to either class depending on initialisation, and both collapse modes were observed.

This suggests, but does not establish, that the original's Table 4 and Figure 6 describe different runs of an initialisation-sensitive model rather than a single set of results. Under such a reading the accuracy of 93.6% and the false-negative rate of 100% would derive from an all-negative run and the true-positive and false-positive rates of 100% from an all-positive one, which would reconcile the reported values as originating from two different runs.

Three considerations bound what can be inferred from this. First, we cannot observe the runs that produced the original's reported values, and no combination of published information distinguishes this account from other explanations, including transcription or tabulation error. Second, the all-positive run we observed terminated at epoch 1, with the best validation loss remaining at epoch 0; no trained epoch improved on the initial validation loss. This run therefore does not establish that an all-positive classifier would persist after successful optimisation. Third, the original also reports a mean squared error of 0.07111, but the published information does not allow us to determine whether this value came from the same run as the reported confusion matrix and rates.

We therefore report the correspondence as an observation rather than as an explanation. What the reproduction establishes is narrower and does not depend on the interpretation above: under the original's own stated definitions, the reported metric values cannot jointly describe the same run as the published confusion matrix. At least one reported value must therefore refer to a different result, contain an error, or arise from a different calculation. Determining which would require access to the original implementation.

### 4.3 Obstacles to Reproduction

The reconstruction required decisions at points where the published description did not determine an implementation. These are worth setting out, both because they bound the confidence of the results above and because the categories they fall into recur beyond this particular study.

**Specifications that cannot be executed as written.** Two of the original's imputation rules depend on one another for their inputs. Rule 4 derives the *STDs* indicator from two individual disease columns, and Rule 5 imputes those same columns by reference to *STDs*. Because all three columns share an identical missingness pattern, each rule requires as input precisely the values the other produces, and no execution order satisfies both (Section 2.2). A reader cannot recover the intended procedure from the text; we adopted the only order under which both rules have defined inputs and recorded the deviation.

**Quantities stated in a form that does not support the stated use.** Rule 3 supplies two conditional proportions of the form  $P(\text{covariate} \mid \text{IUD})$ , whereas imputing the IUD variable requires  $P(\text{IUD} \mid \text{covariate})$ . No base rate is given by which to invert them, and no assignment rule is stated. We evaluated three candidate readings against the records with observed IUD status; the best agreed with 74.9% of them, so approximately a quarter of the 117 imputed values are expected to be incorrect under any available reading. This is the largest single source of uncertainty in the reconstruction.

**Steps performed but not described.** The variable *STDs:HIV* appears in the original's Figure 2 and therefore appears to have been retained through preprocessing, but no rule anywhere in the text addresses it. Input normalisation is a more consequential instance: no scaling step is described, yet without one the network begins training in a strongly saturated regime with heavily attenuated gradients (Section 2.4). Both steps are plausible components of the pipeline that produced the published results, but neither procedure is explicitly recoverable from the paper.

**Internally inconsistent statements.** The original specifies squared error in Algorithm 1 and reports cross-entropy in its results section; states that twelve features were used while tabulating ten; states one training epoch while its training figure shows nine; and reports three mutually incompatible sets of classification metrics. Each of these required us to select a reading.

**Undetermined parameters.** Random Forest hyperparameters, all random seeds, and whether the data partition was stratified are not reported. These omissions prevent exact reconstruction of any particular original run. The random initialisation is especially consequential here, since our 30-seed analysis produced qualitatively different collapse modes (Section 3.4); the unreported Random Forest settings appear less consequential for reproducing the published feature ranking, since the resulting ranking agreed closely with the published one under our default reconstruction.

**Consequences.** These obstacles had markedly different effects. The alternative interpretations we evaluated did not change the outcome: median true-positive rate remained 0.000 and median accuracy remained at the majority-class baseline across all twelve configurations (Section 3.5), so the central qualitative conclusion is recoverable without resolving these internal inconsistencies. The undocumented normalisation step was by contrast decisive, in that omitting it changes the optimisation regime substantially. The circular imputation rules and the IUD specification affect the reconstructed dataset itself, and their effects cannot be fully quantified from the published information.

It is worth noting which of these mattered least. The inconsistencies most visible on reading the paper—the feature count, the loss function, the epoch count—did not alter the qualitative conclusion under any interpretation we tested. Among the ambiguities we evaluated, the step with the clearest effect on optimisation behaviour was one the paper does not mention. This asymmetry suggests that completeness of description bears more directly on reproducibility than internal consistency: a reader can adjudicate between two stated alternatives, but cannot recover a step that was never stated.

#### 4.4 Evaluation Under Class Imbalance

The dataset used here contains 55 positive cases among 858 records, a positive prevalence of 6.41%. At that prevalence a classifier that predicts the negative class for every observation attains 93.59% accuracy, and this is what the original reports.

The difficulty is not that accuracy is a poor metric in general, but that on imbalanced data it is uninterpretable without a reference point. The reported figure of 93.6% is compatible with a model that has learned a useful relationship and with one that ignores its inputs entirely, and nothing in the number itself distinguishes the two. Reporting the majority-class baseline alongside resolves this at no cost: the difference between the two values is the quantity that carries information about the model, and here it is zero on every partition (Section 3.2).

Three further reporting practices would have made the behaviour visible in this case. Sensitivity and specificity, or an explicit confusion matrix, distinguish performance on the two classes—the original does publish a confusion matrix, and it is from that matrix rather than from the reported metrics that the model’s behaviour can be determined. Threshold-free metrics separate ranking ability from classification: here they show weak discrimination that the reported operating point does not exploit (Section 3.6). Repeated runs establish whether a reported result is characteristic or incidental; a single run of this pipeline could have produced either collapse mode (Section 3.4). The choice of decision threshold deserves particular attention. A threshold of 0.5 is a default rather than a principled choice, and under strong imbalance it is often the wrong one. In our reproduction the threshold maximising Youden’s  $J$  on the test partition was

0.061, close to the positive prevalence, at which the classifier recovered five of eight positive cases at a false-positive rate of 0.331. That operating point is not obviously appropriate for deployment either, but the difference between it and the reported one is substantial, and the original does not report a threshold or indicate that any alternative was considered.

None of this is specific to the study reproduced here. The combination of a rare positive class, accuracy as the headline metric, no baseline comparison, and an unexamined default threshold has been documented as a source of misleading evaluation in imbalanced-classification literature, and it produces results that are simultaneously reproducible and uninformative. The corrective is inexpensive: a baseline figure alongside every accuracy, a threshold-free metric reported with its reference value, and some indication of variability across runs.

## 4.5 Clinical Implications

The reproduced classifier is not suitable for diagnostic or screening use at the operating point reported. It attains 93.6% accuracy while assigning every observation to the negative class, giving a sensitivity of zero and a false-negative rate of 100%: all 55 biopsy-positive records in the dataset are classified as negative.

The asymmetry of costs in screening makes this more consequential than the accuracy figure suggests. A model with sensitivity zero provides no benefit over a rule that assigns every record to the negative class, regardless of its headline accuracy.

The reported false-positive rate warrants comment for the same reason. A false-positive rate of zero appears favourable in isolation, but it is attained here trivially, by never predicting the positive class, and is achieved together with a false-negative rate of 100%. In a screening context the two quantities cannot be assessed independently: any classifier can drive either to zero by ignoring the corresponding class, and a low false-positive rate is informative only alongside a sensitivity that is not zero.

This does not imply that the underlying data are uninformative or that the approach is without value. The continuous outputs discriminate weakly above the random-ranking reference (Section 3.6), and at a threshold matched to the positive prevalence the same model recovers five of eight positive cases in the test partition, albeit at a false-positive rate of 0.331. What the reproduction establishes is narrower: that the classifier as reported, at the threshold reported, detects no positive cases, and that its headline accuracy reflects the class distribution rather than any diagnostic capability.

## 4.6 Limitations

Several limitations bound the conclusions above.

**The software hypothesis is unverified.** Our treatment of input normalisation, the stopping rule and the alternative architecture and initialisation variants rests on the hypothesis that the original was implemented in MATLAB's Neural Network Toolbox (Section 2.4). This is inferred from the reported training algorithm, network size and the form of the published training figure, and is supported by the agreement between our stopping behaviour and the original's, but it is not established. If the original used different software, the defaults we adopted may not correspond to what was applied.

**No individual original run can be reproduced.** Random seeds, Random Forest hyperparameters and the stratification of the data partition are not reported, so our results describe the distribution of outcomes the pipeline produces rather than the particular run the original reports. Given the variation across initialisations documented in Section 3.4, this is a material limitation on any comparison at the level of a single number.

**The reconstructed dataset is not certain to match the original.** Approximately a quarter of the 117 imputed IUD values are expected to be incorrect under the best available reading of Rule 3 (Section 2.2), and the circular dependency between Rules 4 and 5 was resolved by a choice the original does not specify. The dataset on which our model was trained therefore differs from the original's by an unknown amount. The close agreement of the feature-importance ranking ( $\rho = 0.975$ ) suggests the difference is small, but does not bound it.

**Feature selection precedes the data partition.** We retained the pipeline order given in the original, in which the Random Forest ranking is computed on all 858 records including those later allocated to the test partition (Section 2.3). This constitutes information leakage from the test partition, and the test metrics we report are correspondingly optimistic; an unbiased evaluation would therefore be expected to perform no better. That variant is not itself a reproduction of the original, which specifies the leaky ordering.

**The reported mean squared error is not evaluated.** Because the original specifies squared error in one place and cross-entropy in another, we could not determine which quantity its reported value of 0.07111 refers to, and did not attempt a comparison (Section 2.5).

**Repetition counts are modest.** The stability analysis uses 30 seeds and the configuration grid ten seeds per cell. These are sufficient to establish that the all-negative outcome is typical rather than incidental, but not to characterise the tails of the distribution: the single all-positive run observed gives an estimated frequency with a wide interval, and we do not report confidence intervals on any of the aggregate metrics.

**Scope.** This is a reproduction of one study. The reporting practices discussed in Section 4.4 recur widely, but we have not surveyed the literature systematically and make no quantitative claim about their prevalence.

## 5 Conclusion

We reimplemented the CervDetect pipeline from its published description and reproduced the confusion matrix reported in the original study exactly: 803 true negatives, 55 false negatives, and no true or false positives across all 858 observations. The reproduced classifier assigns every record to the negative class. Its accuracy of 93.6% is the proportion of negative cases in the dataset, obtainable by a classifier that ignores every input variable, and its sensitivity is zero.

The reported metrics are not internally consistent. Under the original's own stated definitions, the values given in its Table 4 cannot jointly describe the same run as its published confusion matrix, and at least one reported figure must refer to a different result, contain an error, or arise from a different calculation.

The behaviour we observe is not an artefact of our reconstruction. It persists across 30 random initialisations and across all twelve interpretations of the original's ambiguous implementation details that we evaluated. Among the ambiguities we evaluated, the step with the clearest effect on optimisation behaviour was one the paper does not describe at all.

The reproduction nonetheless has continuing use. The implementation is openly available and fully specified, including the resolutions adopted for each ambiguity in the original description, and provides a reference against which alternative preprocessing, class-imbalance handling, threshold selection or model choices can be compared on this dataset. Its continuous outputs discriminate weakly above the random-ranking reference, so the pipeline is a starting point for such comparisons rather than a dead end.

The reported accuracy is therefore reproducible but uninformative, and the study's detection claims are not supported by its own results. Reporting a majority-class baseline alongside accuracy, a threshold-free metric with its reference value, and some measure of variability across runs would have made this visible in the original.

## Declaration

**Conflict of Interest.** The authors declare no conflict of interest.

**Funding.** This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

**Author Contribution.** P. Sahu: conceptualisation, methodology, software, formal analysis, investigation, visualisation, writing — original draft. P. Sudele: validation, writing — review and editing.

**Data Availability.** This study uses the Cervical Cancer (Risk Factors) dataset, publicly available from the UCI Machine Learning Repository [2]. No new data were generated.

**Code Availability.** The complete implementation, including the preprocessing rules, the resolutions adopted for each ambiguity in the original description, the sensitivity variants and the code generating each figure and table, is available at <https://github.com/Prakhar-Sahu26/cervdetect-rescience>.

**Use of Generative AI.** Generative AI tools (OpenAI GPT and Anthropic Claude) were used during the preparation of this work for debugging code, checking factual details and reviewing draft text. All analyses, implementation decisions and reported results were produced and verified by the authors, who take full responsibility for the content of this manuscript.

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